

Aravind Gudikandula

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EDUCATION

- **San Jose State University** San Jose, CA
Master of Science, Software Engineering (Computer Science) Dec 2025
- **Malla Reddy College of Engineering and Technology** Hyderabad, India
Bachelor of Technology, Information Technology June 2023

SKILLS SUMMARY

- **Frontend Development:** React, Angular, Vue.js, JavaScript, TypeScript, HTML5, CSS3, REST APIs, Responsive Design
- **Backend Development:** Java, Python, C++, C, Golang, Ruby, Node.js, Express, SQL, REST APIs, Networking, Data Structures & Algorithms
- **Gen AI & LLM Technologies:** GPT-4, Gemini API, OpenAI, Copilot, LLaMA, RAG, LangChain, Prompt Engineering, LLM Fine-tuning, LLM Integration, Data Analytics, Data Visualisation
- **ML & MLOps:** PyTorch, TensorFlow, Scikit-learn, MLOps Pipelines, Model Deployment, YOLOv8, OpenCV, Model Fine-tuning, NLP, Neural Networks, Data Mining, Data Processing, Statistics, Data Science
- **Cloud & Data Platforms:** AWS (EC2, S3, Lambda, RDS, CloudFormation, DynamoDB, ECS, SQS), GCP (Cloud Run/Build), Azure, MySQL, PostgreSQL, MongoDB, PowerBI
- **Testing & Automation:** Selenium, JUnit, Pytest, Test Automation, Groovy, REST API Testing, Unit Testing, Integration Testing
- **DevOps Tools:** Docker, Kubernetes, Terraform, CI/CD, Jenkins, Git, Linux, JIRA, Confluence, Agile/SDLC
- **Professional Skills:** Technical Skills, Authorization, Development, Communication Skills, Leadership, Innovation

EXPERIENCE

- **Hitachi Vantara** Remote
Associate Software Engineer Jan 2023 - Dec 2023
 - Developed full-stack enterprise solutions using Java, Python, and JavaScript frameworks (React, Angular), collaborating in Agile teams to deliver features that improved system latency by 35% and processed 1M+ daily requests.
 - Engineered RESTful APIs and microservices with Flask and Spring Boot, conducting comprehensive code reviews and implementing automated testing using JUnit and Pytest that increased code coverage from 65% to 90%.
 - Built test automation frameworks using Selenium and Python, reducing bug detection time by 40% and accelerating CI/CD pipeline deployment cycles while maintaining 99.9% application uptime.
 - Collaborated with cross-functional teams to document software designs, analyze root causes of production defects, and communicate project status to stakeholders, supporting team delivery of multiple high-priority releases.
- **F13 Technologies** Remote
AWS Intern Feb 2022 - Nov 2022
 - Designed and deployed cloud infrastructure on AWS (EC2, S3, Lambda, RDS, CloudFormation) supporting 5+ scalable applications with 99.8% availability, leveraging infrastructure-as-code practices.
 - Automated deployment workflows and CI/CD pipelines using CloudFormation templates and Python scripts, decreasing deployment time by 30% and minimizing production incidents through repeatable automation.
 - Analyzed cloud resource utilization with CloudWatch and Cost Explorer, identifying optimization opportunities that delivered 18% cost savings in monthly AWS expenditure while maintaining performance targets.
 - Migrated legacy applications to cloud-native architectures, improving system scalability and reducing infrastructure maintenance overhead by 25% through systematic problem-solving and technical documentation.

PROJECTS

- **Resolve AI – AI-Powered Visual Repair Assistant** Dec 2025
Skills: React 19, TypeScript, Express.js, Gemini 2.5 Flash, Supabase, Google Cloud Run, Docker
 - Built full-stack AI diagnostic application using React 19 and Express.js, integrating Gemini 2.5 Flash for multimodal analysis with 90%+ reliability and real-time WebRTC media capture.
 - Deployed containerized application to Google Cloud Run with multi-stage Docker builds, integrated Supabase PostgreSQL for persistent scan history and enabled auto-scaling capabilities.
- **Medical Domain RAG System with LLM Integration** May 2024
Skills: BioMistral, Qdrant, LangChain, FastAPI, Llama.cpp, PubMedBERT, Python
 - Established open-source RAG system leveraging BioMistral (7B) and PubMedBERT, achieving 85% relevance accuracy on healthcare queries with local LLM deployment for data privacy.
 - Enhanced embedding pipeline using Qdrant vector database and LangChain with FastAPI integration, maintaining sub-second response times while decreasing overhead by 40%.
- **Computer Vision Framework for Smart Vehicles** Jan 2025 - Dec 2025
Skills: Python, PyTorch, YOLOv8, Stable Diffusion, CLIP, FAISS, CUDA
 - Architected production-ready computer vision framework integrating Detection, Search, and Generative Insertion modules, exceeding targets with PSNR of 32.44 dB (29% above) and SSIM of 0.9226.
 - Streamlined ML pipeline implementing batch processing and FP16 mixed precision, achieving 12-24x speedup that accelerated processing from 6 hours to 15 minutes with 22 FPS at 95% accuracy.